Transformations of reciprocal trigonometric functions

1 Consider the function $f(x) = \csc x$, which is defined as the reciprocal function of $\sin x$.

a Describe the values of x where $\sin x = 0$.

b What is true about the function $f(x) = \csc x$ at these x-values?

2 Consider the function $f(x) = \sec x$, which is defined as the reciprocal function of $\cos x$.

a Describe the values of x where $\cos x = 0$.

b What is true about the function $f(x) = \sec x$ at these x-values?

3 Consider the function $f(x) = \cot x$, which is defined as the reciprocal function of $\tan x$.

a Describe the values of x where $\tan x = 0$.

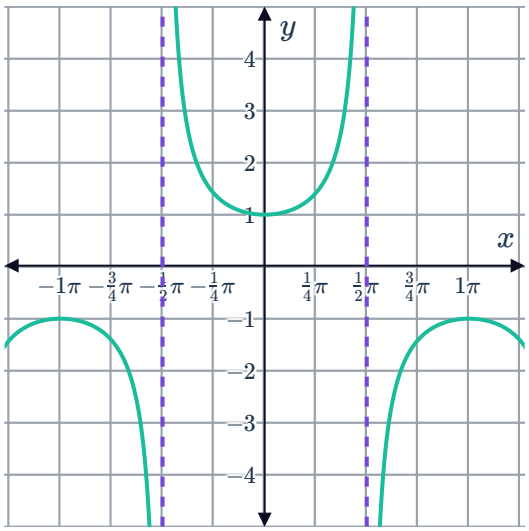
b What is true about the function $f(x) = \cot x$ at these x-values?

4 Consider the graph of $y = \sec x$ and its first local minimum for $x \geq 0$ at $(0, 1)$.
State the coordinates of the first local minimum of each of the given functions for $x \geq 0$:

a $y = \sec 3x$

b $y = \sec\left(x - \frac{\pi}{4}\right)$

c $y = 5\sec x + 2$

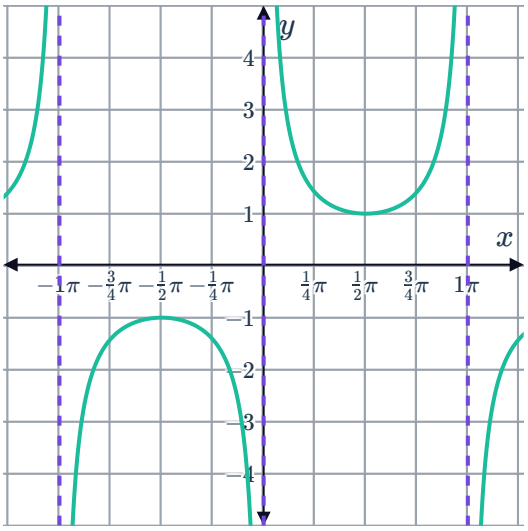


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5 Consider the graph of $y = \csc x$ and its first local minimum for $x \geq 0$ at $\left(\frac{\pi}{2}, 1\right)$.
State the coordinates of the first local minimum of each of the given functions for $x \geq 0$:

a $y = 5\csc x$ **b** $y = -5\csc x$

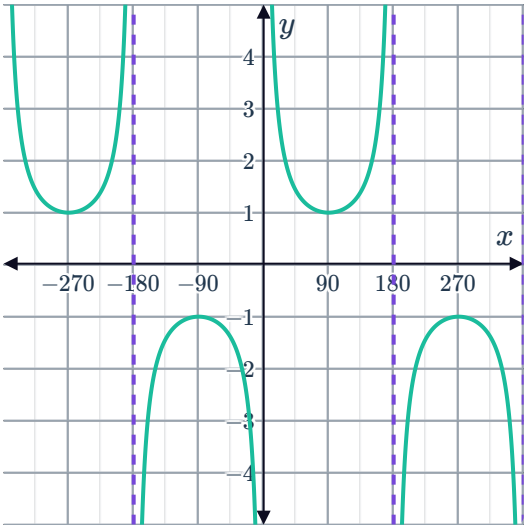
c $y = \csc x + 2$



6 Consider the graph of the function $y = \csc x$:

a Sketch the graph of $y = \csc(x + 90^\circ)$.

b Now, sketch the graph of $y = -2\csc(x + 90^\circ)$.



7 Consider the function $y = \csc x$.

a Sketch the graph of the function $y = \csc\left(x + \frac{3\pi}{4}\right)$ for $-\pi \leq x \leq \pi$.

b Now, sketch the graph of the function $y = \frac{3}{2}\csc\left(x + \frac{3\pi}{4}\right)$ for $-\pi \leq x \leq \pi$.

8 Sketch the graph of the following functions for $-2\pi \leq x \leq 2\pi$:

a $y = 2\sec\left(2x - \frac{\pi}{2}\right) - 2$ **b** $y = 3\csc\left(\frac{x}{2} - \pi\right) - 1$

c $y = 2\cot\left(\frac{x}{2} - \frac{\pi}{2}\right) - 1$

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9 Consider the function $y = \cot\left(3\left(x - \frac{\pi}{6}\right)\right)$.

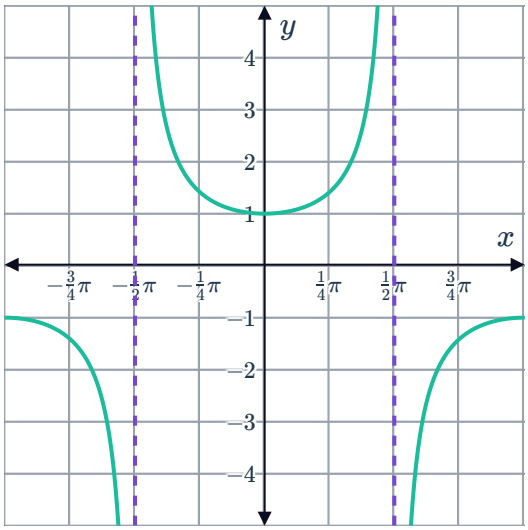
Describe the series of transformations to get the graph of $y = \cot\left(3\left(x - \frac{\pi}{6}\right)\right)$ from $y = \cot x$.

10 Consider the graph of $y = \sec x$.

a Describe the transform from the graph of $y = \sec x$ to create the graph of $y = \sec\left(x - \frac{\pi}{2}\right) - 2$.

b Now, sketch the graph of the function $y = \sec\left(x - \frac{\pi}{2}\right) - 2$ on the same coordinate plane.

c Find the period of $y = \sec\left(x - \frac{\pi}{2}\right) - 2$.



Domain and range of reciprocal trigonometric functions

11 Consider the function $f(x) = \sec x + 3$.

a Describe the transformations of $g(x) = \sec x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Vertical asymptotes

ii Relative minima and maxima

iii Range

iv Domain

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

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12 Consider the function $f(x) = \csc x + 2$.

a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Range

ii Domain

iii Relative minima and maxima

iv Vertical asymptotes

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

13 Consider the function $f(x) = \csc\left(x + \frac{3\pi}{4}\right)$.

a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Range

ii Domain

iii Vertical asymptotes

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

14 Consider the function $f(x) = \sec\left(x - \frac{3\pi}{4}\right)$.

a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Domain

ii Vertical asymptotes

iii Range

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

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15 Consider the function $f(x) = \cot\left(x - \frac{3\pi}{4}\right)$.

a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Domain

ii Range

iii Vertical asymptotes

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

16 Consider the function $f(x) = \csc 4x$.

a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Vertical asymptotes

ii Range

iii Domain

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

17 Consider the function $f(x) = \cot\left(\frac{x}{3}\right)$.

a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Domain

ii Range

iii Vertical asymptotes

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

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18 Consider the function $f(x) = \sec 3x$.

a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Period

ii Domain

iii Range

iv Vertical asymptotes

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

19 Consider the function $f(x) = 3\cot x$.

a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Domain

ii Range

iii Vertical asymptotes

iv Period

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

20 Consider the function $f(x) = 4\sec x$.

a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Vertical asymptotes

ii Period

iii Range

iv Domain

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

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21 Consider the function $f(x) = 5\csc x$.

a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.

b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:

i Period

ii Vertical asymptotes

iii Range

iv Domain

c Now, find the domain of $f(x)$ in interval notation.

d State the range of $f(x)$ in interval notation.

22 Consider the function $f(x) = \csc\left(x + \frac{1}{4}\pi\right) - 6$.

a State the period of the function.

b For what values of x are there vertical asymptotes on the graph of $y = f(x)$?

c Find the coordinates of the relative maxima and relative minima in the domain $(0, 2\pi)$.

d Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.

e Find the domain of $f(x)$ in interval notation.

f Find the range of $f(x)$ in interval notation.

23 Consider the function $f(x) = \cot\left(x + \frac{1}{4}\pi\right) - 6$.

a State the period of the function.

b For what values of x are there vertical asymptotes on the graph of $y = f(x)$?

c Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.

d Find the domain of $f(x)$ in interval notation.

e Find the range of $f(x)$ in interval notation.

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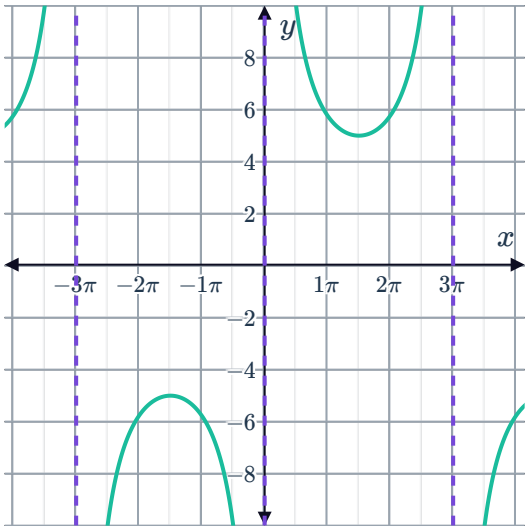
24 For each of the following functions:

i Determine the period of the function.

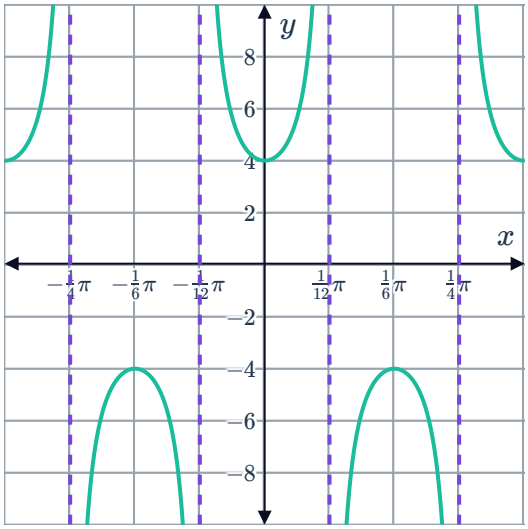
ii Determine the horizontal translation (phase shift) of the function.

iii State the range of the function.

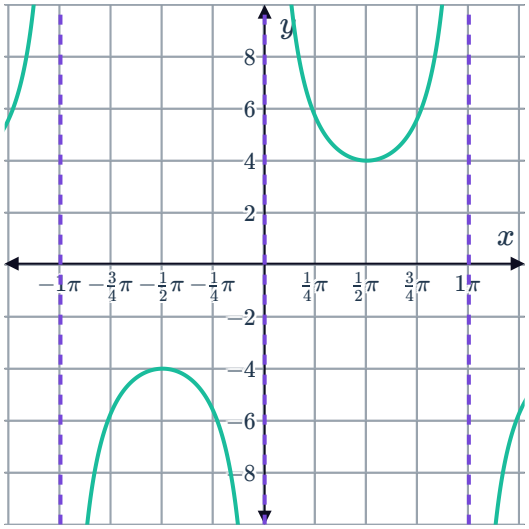
a $y = 5\csc\left(\frac{1}{3}x\right)$



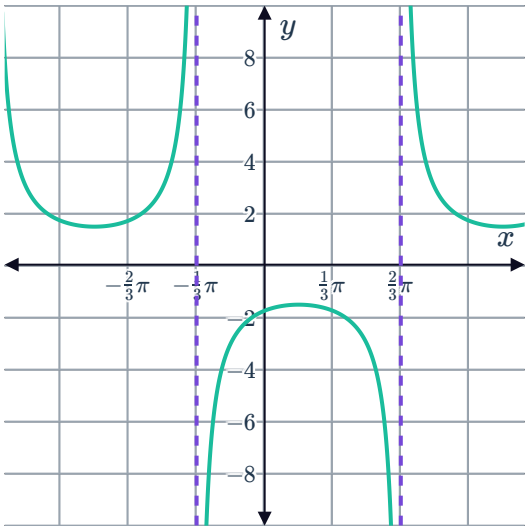
b $y = 4\sec(6x)$



c $y = -4\sec\left(x + \frac{\pi}{2}\right)$



d $y = -\frac{3}{2}\csc\left(x - \frac{\pi}{6}\right)$



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25 Consider the function $f(x) = \sec\left(x + \frac{1}{4}\pi\right) - 3$.

- **a** State the period of the function.
- **b** For what values of x are there vertical asymptotes on the graph of $y = f(x)$?
- **c** Find the coordinates of the relative maxima and relative minima in the domain $(0, 2\pi)$.
- **d** Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.
- **e** Find the domain of $f(x)$ in interval notation.
- **f** Find the range of $f(x)$ in interval notation.

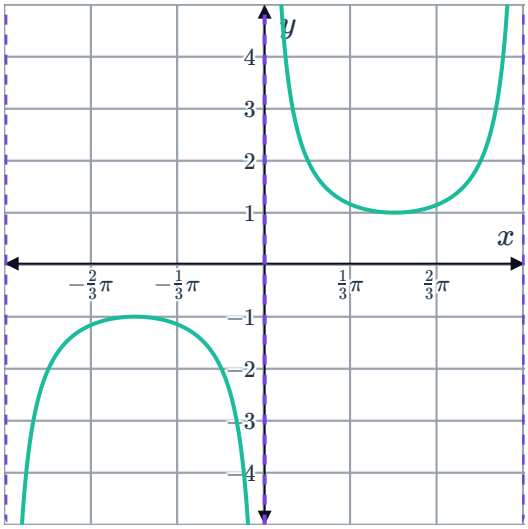
Equations of reciprocal trigonometric functions

26 For each of the following functions, determine the equation of the new function after undergoing the series of transformations:

- **a** The curve $y = \csc x$ is reflected over the x-axis and translated 3 units down.
- **b** The curve $y = \sec x$ is translated $\frac{\pi}{2}$ units to the left, and then reflected over the x-axis.

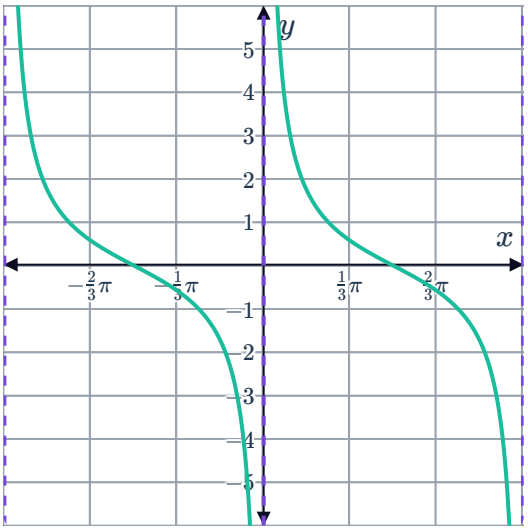
27 The graph of $y = \csc x$ shown is stretched vertically by a scale factor of 2 and translated horizontally $\frac{\pi}{3}$ units to the left.

- **a** Find the equation of the new graph.
- **b** Sketch the graph of the transformed function.



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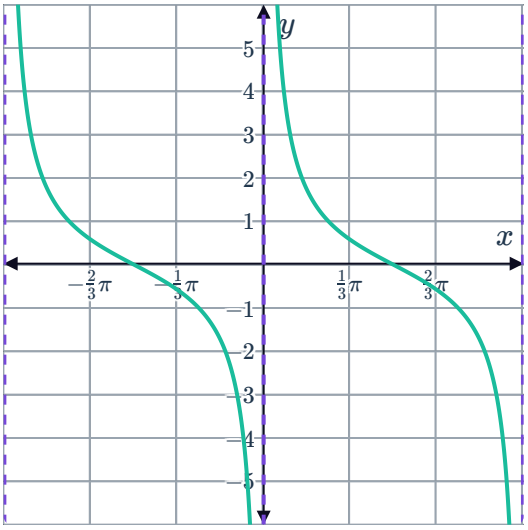
28 The graph of $y = \cot x$ shown is stretched vertically by a scale factor of 5 and translated horizontally $\frac{\pi}{6}$ units to the right.



a Find the equation of the new graph.

b Sketch the graph of the transformed function.

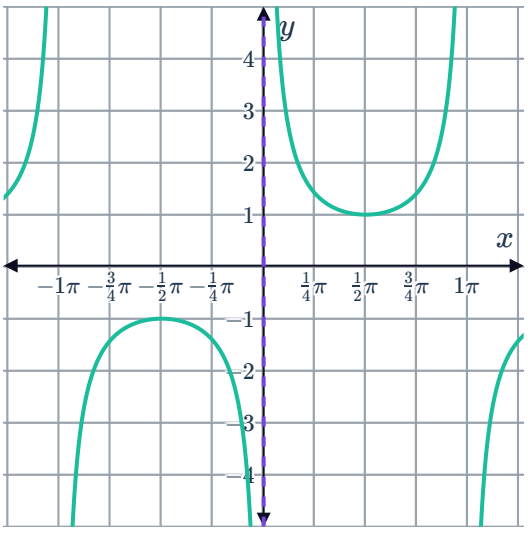
29 The graph of $y = \cot x$ shown is compressed horizontally by a scale factor of $\frac{1}{3}$ and then reflected over the x-axis.



a Find the equation of the new graph.

b Sketch the graph of the transformed function.

30 Consider the following graph:



a Find the equation of the asymptote shown.

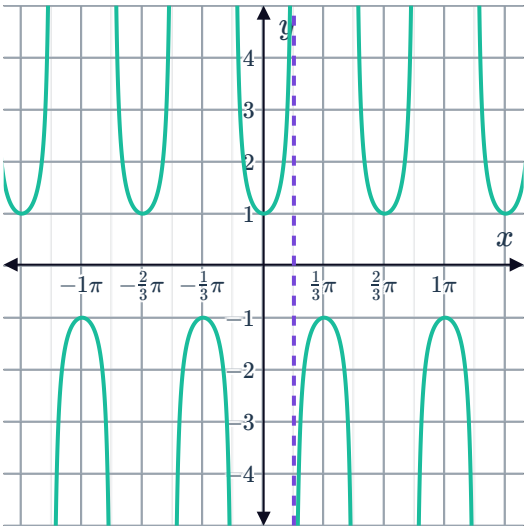
b Which key feature occurs at the point where $x = \frac{\pi}{2}$?

c Find the period of this function.

d Find the equation of this function.

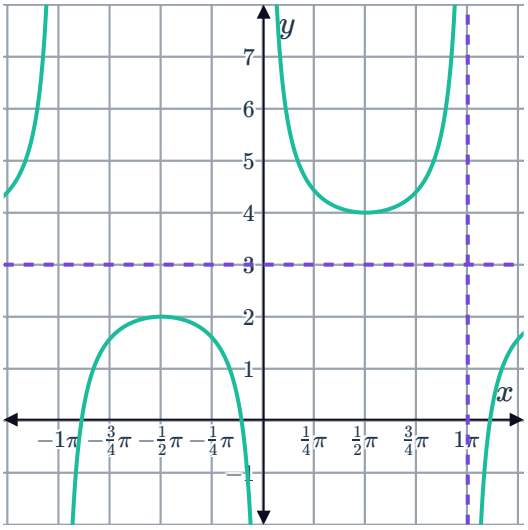
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31 Consider the following graph:



- **a** Find the equation of the asymptote shown.
- **b** Which key feature occurs at the point where $x = \frac{\pi}{3}$?
- **c** Find the period of this function.
- **d** Find the equation of the equation of this function.

32 Consider the following graph:



- **a** Find the equation of the asymptote shown.
- **b** Find the period of this function.
- **c** Find the median value of this function.
- **d** Find the equation of this function.

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